

SayedMorteza Malaekheh

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EDUCATION

University of Texas at Austin (UT Austin)

PhD Candidate (ABD) in Sustainable Systems Engineering

Jan. 2023 – Exp. Dec. 2026

Joint M.Sc. in Economics, conferred May 2026 (full Ph.D. sequence, 12 courses)

California Institute of Technology

Visiting PhD Researcher in Economics and Computer Science Groups

May-Aug. 2024; May-Dec. 2025

Sharif University of Technology

M.Sc. in Water Resources Engineering (Thesis A+; Ranked First)

Sep. 2019 – Feb 2022

Sharif University of Technology

B.Sc. in Civil Engineering with Minor in Economics

Sep. 2015 – Aug. 2019

SELECTED WORK/RESEARCH EXPERIENCE

PhD Data Science Intern | Roblox Corporation

May – Aug. 2026

- Improved the Roblox Moments short-form video recommender system by redesigning ranking objectives to optimize long-term retention instead of short-term engagement.
- Applied double machine learning, causal forests, and deep learning models to petabyte-scale behavioral data, improving long-term-retention ranking by 30%+.
- Designed Bayesian optimization for A/B testing across short- and long-run outcomes to efficiently identify and validate improved recommender-system ranking objectives.
- Built production-ready ML and data pipelines over billions of impression and session events using Spark SQL/Hive and Trino, collaborating with ML, software, and data engineers.

Visiting PhD Researcher | California Institute of Technology

May-Aug. 2024; May-Dec. 2025

- Built a multimodal causal framework integrating deep learning and causal inference.
- Integrated Vision Transformers, CNNs, and autoencoders with R-Learner and Causal Forest.
- Developed a new estimator for conditional average treatment effects with image-based treatments, applied to large multimodal datasets in environmental policy.

Graduate Researcher | Lawrence Berkeley National Laboratory

June 2024 – June 2025

- Designed a nationwide propensity-score matching strategy balancing solar PV adopters and non-adopters.
- Led acquisition of a \$180K Experian credit dataset covering 10M households, 2010–2023.
- Engineered scalable pipelines to clean and analyze 13+ TB of monthly household-level credit data.
- Applied synthetic control and staggered DiD to estimate financial impacts of solar PV adoption.

Quantitative Research Fellow | Quanta Ventures Fund

Jan. – Mar. 2026

- Engineered daily trading strategies for a multi-strategy engine across major ETFs (QQQ, SPY, GLD) using ML and deep learning, reaching Sharpe and Calmar ratios of ~ 2.0 .
- Developed and backtested intraday models on minute-level ETF data to time entry and exit signals.
- Designed alpha signals combining mean reversion, momentum, and breakout logic, with ADX regime filters for adaptability across market environments.

SELECTED PUBLICATIONS (GOOGLE SCHOLAR: 59 CITATIONS. [PROFILE](#))

Manuscripts Under Review

- [1] **Malaekheh, S.**, Castellanos, S., Barbose, G., & O'Shaughnessy, E. **Residential Solar Adoption Leaves Financially Constrained Households at Greater Risk.**

UNDER REVIEW, *Nature Energy*

Working Papers

- [1] **Malaekheh, S.**, Druckenmiller, H., Jerzak, C., & Gkioxari, G. **A Multimodal Causal Framework for Estimating Location-Specific Ecosystem Values at Scale.** [Working Paper]

WORKSHOP VERSION: NEURIPS 2024, TACKLING CLIMATE CHANGE WITH ML (PEER-REVIEWED).

Peer-Reviewed Journal Articles

- [1] **Malaekheh, S.**, French, J., Potts, J., & Castellanos, S. (2026). Multidimensional risk and resilience score: Informing utility investments in vulnerable communities. *Environmental Research Communications*, 8(6), 065009. [\[Article\]](#)
- [2] **Malaekheh, S.**, & Castellanos, S. (2025). Residential solar photovoltaic seeding effects and their unequal impact on diffusion across racial and ethnic groups in the U.S. *Energy Policy*, 206, 114782. [\[Article\]](#)

- [3] **Malaekheh, S.**, Shiva, L., & Safaie, A. (2024). Investigating the economic impact of climate change on agriculture in Iran: Spatial spillovers matter. *Agricultural Economics*, 55(3), 433–453. [\[Article\]](#) [\[Replication\]](#)
- [4] Danesh-Yazdi, M., & **Malaekheh, S.** (2023). Investigating water resources and agricultural economy in the Mahabad study area by hydro-economic modeling. *Iran-Water Resources Research*, 19(2), 87–105. [\[Article\]](#)
- [5] **Malaekheh, S.**, Safaie, A., Shiva, L., & Tabari, H. (2022). Spatio-temporal variation of hydro-climatic variables and extreme indices over Iran based on reanalysis data. *Stochastic Environmental Research and Risk Assessment*, 36, 3725–3752. [\[Article\]](#)

PRESENTATIONS

- 2026** American Solar Energy Society National Solar Conference (ASES, oral), Austin, TX [\[UPCOMING\]](#)
- 2025** U.S. Association for Energy Economics (USAEE, oral), Fort Worth, TX; Workshop in Environmental Economics and Data Science (TWEEDS, Causal AI panel), Portland, OR
- 2024** NeurIPS Workshop on Tackling Climate Change with ML [\[link\]](#); American Geophysical Union (AGU, oral), Washington, DC; Sloan/Berkeley Summer School in Environmental & Energy Economics (Egg Timer), UC Berkeley; Macro Energy Systems (MES), Princeton
- 2023** American Geophysical Union (AGU), San Francisco; Seminar in Water Economics OnLine (SWELL)
- 2022** European Geosciences Union (EGU, oral), Vienna
- 2021** European Geosciences Union (EGU), online

FELLOWSHIPS, & AWARDS

- 2026 **Academy Distinguished Alumni Graduate Fellowship**, UT Austin
- 2025 **Robert Abbasi & Shahnaz Hemmati Graduate Fellowship**, UT Austin
- 2024 **Grad Camp Fellowship**, UC Berkeley & **Travel Grant**, Princeton University
- 2020 **Exchange Program Scholarship**, Sharif Univ. of Technology
- 2019 **Ranked 9th/10,000 (top 0.1%)** — Iranian National Economics Olympiad

SKILLS

- Programming:** Python (PyTorch, Scikit-learn, XGBoost), SQL, R, MATLAB, STATA
- ML/AI:** Recommender Systems, Deep Learning, LLMs (Hugging Face, LangChain, PEFT/LoRA), Bayesian Optimization
- Causal/Experimentation:** A/B Testing, Causal Machine Learning, DiD, DML/R-Learner, Causal Forests, Synthetic Control, Regression Discontinuity, EconML
- Data/Systems:** Spark/PySpark, Hive, Trino, Databricks, AWS, GCP, Docker, Kubernetes, HPC